

A Counterexample to Uniqueness for a Parabolic Min–Max Variational Inequality with Obstacle and Gradient Constraints

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Abstract

We give an explicit counterexample to uniqueness for the Sobolev solution of the min–max variational inequality with obstacle and gradient constraints appearing as Problem A in the work of Bovo, De Angelis, and Issoglio. Thus the uniqueness question in Remark 3.5 of that paper has a negative answer under the stated hypotheses: this is a solution by disproof, not merely partial progress. The construction is one-dimensional and smooth. For the same coefficients and data, the functions

$$u_\lambda(t, x) = \frac{\pi}{2} + \arctan x + \lambda(T - t), \quad 0 \leq \lambda \leq 1,$$

are all Sobolev solutions with quadratic growth and the same terminal condition. Hence there are infinitely many solutions. The mechanism is that the gradient constraint is saturated everywhere, so the inaction set is empty and the free-boundary formulation imposes only a one-sided differential inequality.

1 Introduction and verdict

Bovo, De Angelis, and Issoglio study a zero-sum singular-controller versus stopper game and associate it with a parabolic variational inequality of min–max type with an obstacle constraint and a gradient constraint. Their main theorem identifies the value function as the maximal solution of their Problem A. Remark 3.5 asks whether the solution of Problem A is unique.

The answer is no. The example below satisfies the structural assumptions used for Problem A and produces a continuum of distinct strong Sobolev solutions. Therefore the result obtained here is a complete negative solution of the stated uniqueness problem. It does not contradict the maximal-solution theorem: one can have a distinguished maximal solution while nonmaximal solutions also exist.

2 The variational problem

Fix a finite horizon $T > 0$. Let

$$Q_T := [0, T) \times \mathbb{R}^d, \quad \overline{Q}_T := [0, T] \times \mathbb{R}^d.$$

For $b : \mathbb{R}^d \rightarrow \mathbb{R}^d$ and $\sigma : \mathbb{R}^d \rightarrow \mathbb{R}^{d \times d'}$, set

$$a(x) := \sigma(x)\sigma(x)^\top.$$

The associated generator is

$$(L\varphi)(x) = \frac{1}{2} \operatorname{tr} (a(x) D^2 \varphi(x)) + \langle b(x), \nabla \varphi(x) \rangle.$$

The min–max variational inequality is

$$\min \left\{ \max \left\{ \partial_t u + Lu - ru + h, g - u \right\}, f - \|\nabla u\|_d \right\} = 0 \quad \text{a.e. on } Q_T, \quad (1)$$

with terminal condition

$$u(T, x) = g(T, x), \quad x \in \mathbb{R}^d, \quad (2)$$

and pointwise constraints

$$u \geq g, \quad \|\nabla u\|_d \leq f \quad \text{on } \overline{Q}_T. \quad (3)$$

The companion max–min inequality displayed in the same setting is

$$\max \left\{ \min \left\{ \partial_t u + Lu - ru + h, f - \|\nabla u\|_d \right\}, g - u \right\} = 0 \quad \text{a.e. on } Q_T. \quad (4)$$

The free-boundary formulation of Problem A asks, for $p > d + 2$, for a function

$$u \in W_{\text{loc}}^{1,2,p}(Q_T)$$

with a continuous representative up to the terminal time, with quadratic growth, satisfying the following conditions. Define

$$\mathcal{I} := \{(t, x) \in Q_T : \|\nabla u(t, x)\|_d < f(t, x)\}, \quad \mathcal{C} := \{(t, x) \in Q_T : u(t, x) > g(t, x)\}.$$

Then

$$\partial_t u + Lu - ru = -h \quad \text{on } \mathcal{C} \cap \mathcal{I}, \quad (5)$$

$$\partial_t u + Lu - ru \geq -h \quad \text{a.e. on } \mathcal{C}, \quad (6)$$

$$\partial_t u + Lu - ru \leq -h \quad \text{a.e. on } \mathcal{I}, \quad (7)$$

together with (2), (3), and

$$|u(t, x)| \leq C(1 + \|x\|_d^2) \quad \text{on } \overline{Q}_T \quad (8)$$

for some constant $C > 0$. In the source paper this formulation is stated to be equivalent to the variational inequalities (1) and (4), a.e. on Q_T , for functions in the relevant Sobolev class with quadratic growth. We will verify both formulations directly for the example.

3 The counterexample

Work in dimension $d = d' = 1$. Set

$$b(x) = 0, \quad \sigma(x) = 1, \quad a(x) = 1, \quad r = 0.$$

Thus

$$L = \frac{1}{2} \partial_{xx}.$$

Define time-independent data

$$g(t, x) = g(x) := \frac{\pi}{2} + \arctan x, \quad f(t, x) = f(x) := \frac{1}{1 + x^2}, \quad (9)$$

and

$$h(t, x) = h(x) := 1 + \frac{x}{(1 + x^2)^2}. \quad (10)$$

For $0 \leq \lambda \leq 1$, put

$$u_\lambda(t, x) := g(x) + \lambda(T - t), \quad (t, x) \in \overline{Q}_T. \quad (11)$$

Theorem 1. *For every $\lambda \in [0, 1]$, the function u_λ is a smooth strong Sobolev solution of (1) satisfying the terminal condition, the obstacle constraint, the gradient constraint, and the quadratic-growth bound. It also satisfies the companion max–min equation (4) and the free-boundary formulation (5)–(7). Consequently uniqueness of Sobolev solutions to Problem A is false under the stated assumptions.*

4 Verification of the standing assumptions

We first check that the data lie within the hypotheses under which Problem A is posed.

4.1 The controlled-diffusion coefficients

The functions b and σ are constant, hence continuously differentiable, locally Lipschitz, and of at most linear growth. Moreover $a(x) = 1$. Therefore, for every bounded set $B \subset \mathbb{R}$ and every $\zeta \in \mathbb{R}$,

$$\zeta^2 = \langle \zeta, a(x)\zeta \rangle \geq 1 \cdot |\zeta|^2, \quad x \in B,$$

so local ellipticity holds with ellipticity constant $\theta_B = 1$.

4.2 Regularity and growth of the data

The functions g and h are smooth in x and independent of t . Also

$$f^2(x) = \frac{1}{(1 + x^2)^2}$$

is smooth. Hence $f^2, g \in C_{\text{loc}}^{1,2,\alpha}(\overline{Q}_T)$ and $h \in C_{\text{loc}}^{0,1,\alpha}(\overline{Q}_T)$ for every $\alpha \in (0, 1)$, in the usual parabolic notation.

The function f is independent of time, so $t \mapsto f(t, x)$ is non-increasing for each x . Moreover $0 < f(x) \leq 1$, hence the polynomial-growth upper bound on f is immediate. The functions g and h are also independent of time, so the one-sided time-growth assumptions on g and h hold with any positive constant.

We next show nonnegativity. Since $\arctan x \in (-\pi/2, \pi/2)$,

$$0 < g(x) < \pi.$$

For $x \geq 0$, the function $x(1 + x^2)^{-2}$ has derivative

$$\frac{1 - 3x^2}{(1 + x^2)^3},$$

so its maximum on $[0, \infty)$ occurs at $x = 1/\sqrt{3}$ and equals

$$\frac{9}{16\sqrt{3}}.$$

By oddness of $x(1+x^2)^{-2}$,

$$\left| \frac{x}{(1+x^2)^2} \right| \leq \frac{9}{16\sqrt{3}} < 1. \quad (12)$$

Therefore

$$h(x) \geq 1 - \frac{9}{16\sqrt{3}} > 0.$$

In particular f, g, h take values in $[0, \infty)$, and $g+h$ is bounded, hence satisfies the required quadratic-growth bound.

4.3 Compatibility of the obstacle with the gradient cap

A direct differentiation gives

$$g'(x) = \frac{1}{1+x^2} = f(x). \quad (13)$$

Thus

$$|\nabla g(t, x)| = f(t, x)$$

for all (t, x) . The compatibility condition $|\nabla g| \leq f$ is therefore satisfied, with equality everywhere.

Finally,

$$g''(x) = -\frac{2x}{(1+x^2)^2}, \quad Lg(x) = \frac{1}{2}g''(x) = -\frac{x}{(1+x^2)^2}.$$

Since $r = 0$ and $\partial_t g = 0$, the lower-bound quantity from the source-paper assumptions is

$$h + \partial_t g + Lg - rg = 1 + \frac{x}{(1+x^2)^2} - \frac{x}{(1+x^2)^2} = 1.$$

This is bounded below, for instance by -1 .

5 Verification of the variational inequalities

Fix $\lambda \in [0, 1]$. From (11) and (13),

$$\partial_x u_\lambda(t, x) = g'(x) = f(x) > 0. \quad (14)$$

Hence

$$|\partial_x u_\lambda(t, x)| = f(x), \quad f(x) - |\partial_x u_\lambda(t, x)| = 0. \quad (15)$$

Moreover,

$$u_\lambda(t, x) - g(x) = \lambda(T - t) \geq 0, \quad g(x) - u_\lambda(t, x) = -\lambda(T - t) \leq 0, \quad (16)$$

and

$$u_\lambda(T, x) = g(x). \quad (17)$$

Thus the obstacle, gradient, and terminal constraints hold.

It remains to compute the differential expression. Since $\partial_t u_\lambda = -\lambda$ and $Lu_\lambda = Lg$, we obtain

$$\begin{aligned} \partial_t u_\lambda + Lu_\lambda - ru_\lambda + h &= -\lambda - \frac{x}{(1+x^2)^2} + 1 + \frac{x}{(1+x^2)^2} \\ &= 1 - \lambda. \end{aligned} \quad (18)$$

Because $0 \leq \lambda \leq 1$, this quantity is nonnegative. Combining (15), (16), and (18), for all $(t, x) \in Q_T$ we have

$$\begin{aligned} & \min \left\{ \max \left\{ \partial_t u_\lambda + Lu_\lambda - ru_\lambda + h, g - u_\lambda \right\}, f - |\partial_x u_\lambda| \right\} \\ &= \min \left\{ \max \left\{ 1 - \lambda, -\lambda(T - t) \right\}, 0 \right\} = 0. \end{aligned}$$

This proves (1) pointwise, hence a.e.

The companion max–min equation also holds pointwise:

$$\begin{aligned} & \max \left\{ \min \left\{ \partial_t u_\lambda + Lu_\lambda - ru_\lambda + h, f - |\partial_x u_\lambda| \right\}, g - u_\lambda \right\} \\ &= \max \left\{ \min \left\{ 1 - \lambda, 0 \right\}, -\lambda(T - t) \right\} = 0. \end{aligned}$$

Finally, u_λ is smooth on $\overline{Q}_T$ and bounded by a constant depending only on T , since $0 < g < \pi$ and $0 \leq \lambda(T - t) \leq T$. Hence

$$u_\lambda \in W_{\text{loc}}^{1,2,p}(\overline{Q}_T)$$

for every finite p , and the quadratic-growth bound is satisfied.

6 Verification of the free-boundary formulation

For u_λ , the inaction set is

$$\mathcal{I}_\lambda := \{(t, x) \in Q_T : |\partial_x u_\lambda(t, x)| < f(x)\}.$$

By (15),

$$\mathcal{I}_\lambda = \emptyset. \tag{19}$$

The continuation set is

$$\mathcal{C}_\lambda := \{(t, x) \in Q_T : u_\lambda(t, x) > g(x)\}.$$

If $\lambda = 0$, then $\mathcal{C}_0 = \emptyset$. If $0 < \lambda \leq 1$, then $T - t > 0$ for every $(t, x) \in Q_T$, so

$$\mathcal{C}_\lambda = Q_T. \tag{20}$$

Because $\mathcal{I}_\lambda$ is empty, the equality on $\mathcal{C}_\lambda \cap \mathcal{I}_\lambda$ and the inequality on $\mathcal{I}_\lambda$ are vacuous. The only nonvacuous free-boundary differential condition occurs when $\lambda > 0$, where $\mathcal{C}_\lambda = Q_T$, and it is

$$\partial_t u_\lambda + Lu_\lambda - ru_\lambda \geq -h.$$

This is equivalent to

$$\partial_t u_\lambda + Lu_\lambda - ru_\lambda + h = 1 - \lambda \geq 0,$$

which holds exactly for $\lambda \in [0, 1]$. Thus u_λ satisfies Problem A in the free-boundary sense as well.

7 Nonuniqueness and scope

Taking $\lambda = 0$ and $\lambda = 1$ gives two distinct solutions:

$$u_0(t, x) = \frac{\pi}{2} + \arctan x, \quad u_1(t, x) = \frac{\pi}{2} + \arctan x + T - t.$$

Since $T > 0$, these are not equal on $[0, T) \times \mathbb{R}$. In fact, the entire one-parameter family $\{u_\lambda : 0 \leq \lambda \leq 1\}$ consists of solutions.

The example is deliberately degenerate in one structural sense: the gradient cap is saturated everywhere. Consequently the inaction set is empty. This is not excluded by the assumptions of Problem A, because the source paper assumes only $|\nabla g| \leq f$, not strict inequality. Thus the example genuinely disproves uniqueness as stated. It would not, by itself, answer a stronger problem in which one imposed a strict compatibility condition such as $|\nabla g| < f$ on a substantial set, or required the inaction set to be nonempty.

8 Conclusion

The uniqueness question for Problem A has a negative answer under the stated hypotheses. The construction gives smooth, bounded, strong Sobolev solutions satisfying all constraints and both displayed variational inequalities. Therefore the contribution is a complete counterexample to uniqueness, not merely partial progress. The counterexample is also consistent with the theorem that the game value is the maximal solution: maximality does not imply uniqueness among all Sobolev solutions.

References

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